



Design and Analysis of Algorithms

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DESIGN AND ANALYSIS OF ALGORITHMS

Performance Analysis Vs Performance Measurement

Slides courtesy of **Anany Levitin**

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Performance Evaluation of Algorithm

- Performance Analysis
 - Machine Independent
 - Prior Evaluation
- Performance Measurement
 - Machine Dependent
 - Posterior Evaluation



Design and Analysis of Algorithms

Performance Analysis of Sequential search :Worst Case



ALGORITHM SequentialSearch($A[0..n-1]$, K)

//Searches for a given value in a given array by sequential search

//Input: An array $A[0..n-1]$ and a search key K

//Output: Returns the index of the first element of A that matches K or -1 if there are no matching elements

$i \leftarrow 0$

while $i < n$ and $A[i] \neq K$ do

$i \leftarrow i + 1$

if $i < n$ // $A[i] = K$

 return i

else

 return -1

Basic operation: $A[i] \neq K$

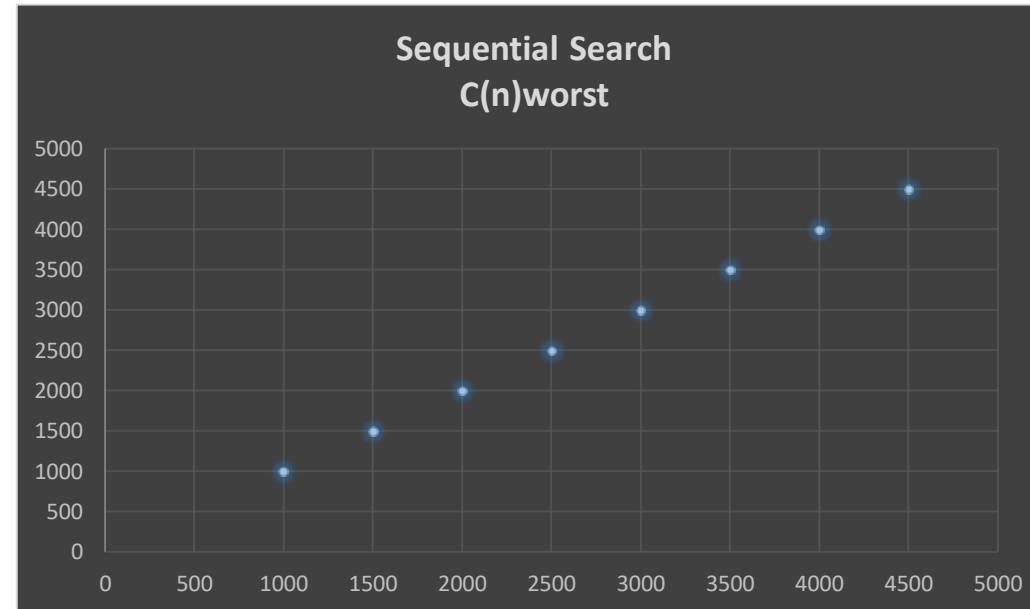
Basic operation count: n

Time Complexity: $T(n) \in O(n)$

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Performance Analysis of Sequential Search

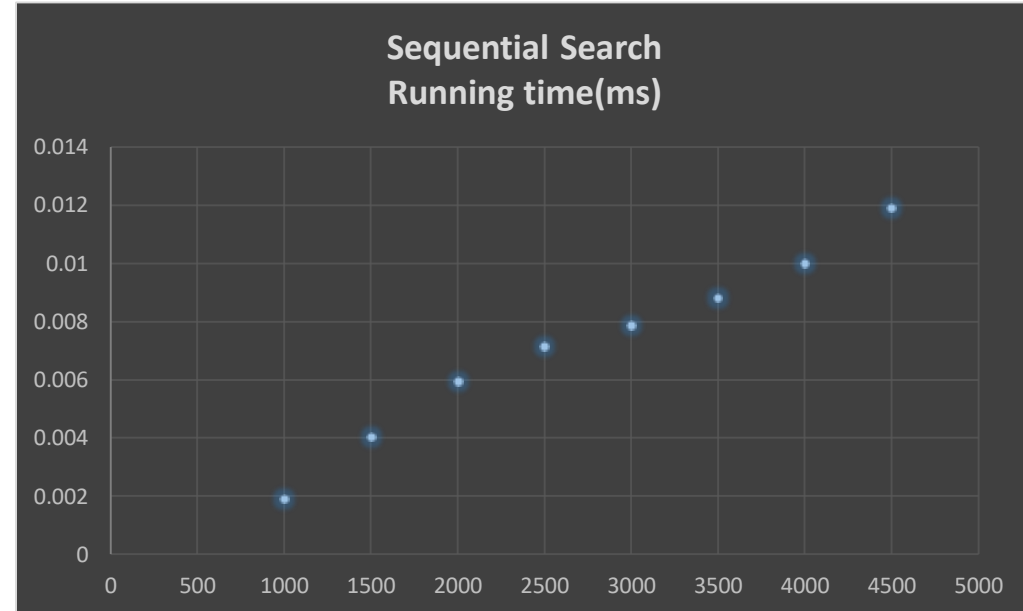
Input Size	Sequential Search $C(n)_{\text{worst}}$
1000	1000
1500	1500
2000	2000
2500	2500
3000	3000
3500	3500
4000	4000
4500	4500



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Performance Measurement of Sequential Search

Input Size	Sequential Search Actual Running Time(ms)
1000	0.001907
1500	0.004053
2000	0.00596
2500	0.007153
3000	0.007868
3500	0.008821
4000	0.010014
4500	0.011921





THANK YOU

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